3rd International Conference and Workshops on Recent Advances and Innovations in Engineering, 22-25 November 2018
(IEEE Conference Record # 43534)

Analyzing the Effect of Partial Shading on Performance of Grid Connected Solar PV System

Anjali
Dept. Electrical Engineering
Arya Institute of Engineering and Technology, Jaipur,India
Anjali.piet@gmail.com

Raj Kumar Kaushik
Dept. Electrical Engineering
Arya Institute of Engineering and Technology, Jaipur,India
Rajkaushik1812@gmail.com

Deepak Sharma
Dept. Electrical Engineering
Arya Institute of Engineering and Technology, Jaipur,India
Deepak.sharma581@gmail.com

***Abstract*—Photovoltaic power is the most rising renewable energy source. Photovoltaic modules have changing properties regarding irradiance and temperature. So as to use the maximum power from the PV module, a maximum power point tracking algorithm is utilized. These algorithms are prepared by the dynamic qualities of the PV modules. Photovoltaic module clusters have unexpected conduct in comparison to individual Photovoltaic modules under halfway shading conditions. Under halfway shading conditions, the Photovoltaic cluster qualities have various pinnacles and the maximum power point algorithm should be prepared to discover the crest with the best power. Checking Photovoltaic cluster qualities under typical irradiance and halfway shading is fundamental to remove maximum power from the PV exhibit. In this paper, the different qualities of photovoltaic exhibits under halfway protecting conditions are examined, and a model for choosing the trademark locales of the cluster is proposed to follow the worldwide maximum power point.**

Keywords*- *Photovoltaic Array, Partial shading condition, global MPP, local MPP, shading patterns

I. INTRODUCTION

The expansion in worldwide energy request and increment consideration regarding natural issues have prompted the investigation of renewable energy sources, for example, solar and wind. Albeit photovoltaic frameworks are progressively utilized in an assortments of uses, the high establishment cost and low transformation productivity of photovoltaic modules are real deterrents to the utilization of photovoltaic power sources [1, 2]. In this way, investigate on Photovoltaic power age frameworks is by and large effectively elevated so as to limit these disservices.

So as to adequately use PV power, the Maximum Power Point Tracker (MPPT) is usually utilized with DC-DC converters [3, 4]. Lift converters are broadly utilized [5] as MPPTs to accomplish higher yield voltages and diminish the quantity of boards in an arrangement string. The primary objective of the MPPT is to guarantee that the framework dependably draws maximum power from the exhibit. In any case, because of changes in natural conditions, for example, solar radiation and temperature, the maximum power point (MPP) in the P-V trademark bend shifts nonlinearly with these conditions, in this manner testing the tracking algorithm [6-8]. Different MPPT plans, for example, unsettling influences and perceptions [9, 10], steady conductance [11, 12], cut off [13] and open circuit voltage [14] have been settled so as to work the PV exhibit on various stages in the MPP. Under conditions. These plans are effectvly work under uniform presentation condition where just a single pinnacle shows up at the MPP voltage of the exhibit [15].

The MPPT issue of photovoltaic exhibits working under non-uniform daylight conditions [20-26] has been tended to in the writing. Genuine maximum power point. The tracking strategy [20] first identifies changes in PV voltage and current to distinguish the event of halfway tinge. At that point, change the working point as indicated by a foreordained straight capacity, and after that change the daily practice.

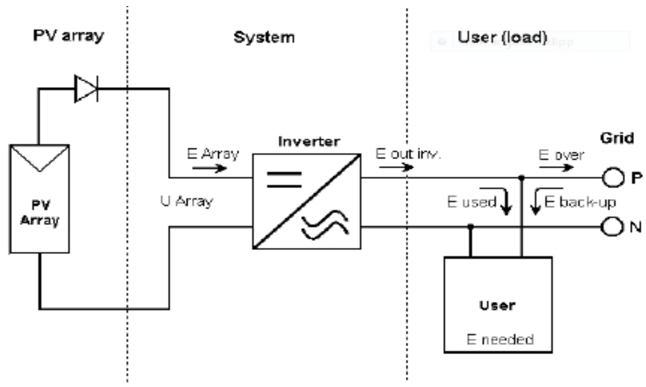

Fig. 1.Grid Connected Solar PV System

Photovoltaic systems consist of photovoltaic arrays, inverters and other system balance (BoS) components such as mounting systems, charge controllers, cables, transformers and suitable accumulators. The choice of components depends on the system and must be chosen separately for each project. Figure 1.1 shows an example of a grid-connected solar system.

II. SYSTEM MODEL

Single and double diode PV cell models [4, 5] have been generally used to anticipate the conduct of PV cells, modules, short strings and little scale exhibits. As of late, a few papers still spotlight on the technique for removing parameters [6-8]. The adaptable granularity is too little to even consider extending to the PV cell level, which isn't viable for handling long PV module strings and substantial scale PV exhibits. This paper utilizes a Photovoltaic module dependent on a piecewise straight full scale model of Psim

• 978-1-5386-4525-3/18/$31.00 ©2018 IEEE

to reproduce a substantial scale halfway shadowed PV exhibit. A large scale demonstrate is a lot of express direct conditions. Along these lines, contrasted and the conventional model, it exhibits a one of a kind arrangement that is quick, exact and solid. Conventional models must explain the nonlinear understood conditions with exponential terms by Newton emphasis, which has the drawbacks of computational intricacy and union issues. Throughout the years, a few reenactment strategies dependent on various programming stages have been created to improve the comprehension and forecast of V-I and V-P qualities of PV cells, models, strings and PV clusters [2, 3, 9, 10]. Walker [9] proposed a PV module display dependent on MATLAB to think about the impacts of its attributes on temperature and occurrence irradiance. Be that as it may, the model does not consider the impact of shadows on PV attributes [3]. Patel and Agarwal [3] have made huge advances in MATLAB-based displaying to contemplate the impacts of incomplete shading on the qualities of photovoltaic exhibits. It very well may be utilized to consider the impacts of temperature and episode irradiance changes and additionally shifting shadow designs. Maki and Valkealahti [10] contemplated the power misfortune in short and little scale varieties of 18 photovoltaic modules dependent on the MATLAB stage. Notwithstanding, these simple strategies utilize a solitary diode PV cell demonstrate that experiences computational multifaceted nature and combination issues.To use PVSYST and concentrate the execution capability of photovoltaic and solar warm frameworks, light information and exact worldwide temperatures are basic. In PVSYST, both breeze speed and diffuse radiation are discretionary.

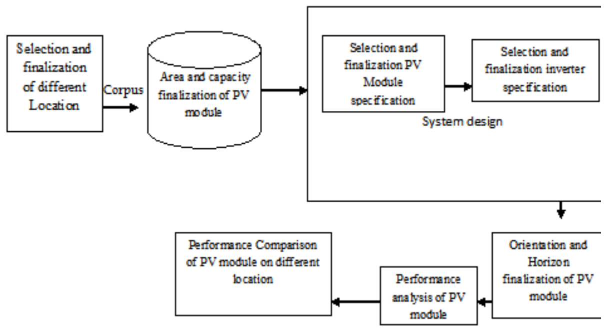

Fig. 2. Architecture of Grid Connected Solar PV System

PVSYST utilizes the irradiance data imported in PVSYST to make unpredictable markers utilizing imported useful climate data. Except if generally expressed, the data in this section is the PVSYST Contextual Help Manual. The design and reenactment of system associated solar PV frameworks was broadened utilizing the accompanying data sets.

Meteological Data-Meteonorm programming gives month to month climate data to each point on the planet. They likewise utilize a stochastic model to give time sensitive data dependent on engineered age. On the off chance that there is a station in a given setting, Meteonorm will utilize interjection between the three stations. Satellite data from five geostationary satellites was utilized as an enhancement when soil data was poor. 8 km level goals. Soil estimation vulnerability ranges from 1% to 10% (Meteonorm results [46]) and satellite data ranges from 3% to 4% (low scope). For ground insertion, the vulnerability is 1% over a separation of 2 km, 6% to 100 km, and 8% for separations more prominent than 2000 km. Utilize the Perez model to figure flat diffuse enlightenment to isolate the worldwide pillar and diffuse segments

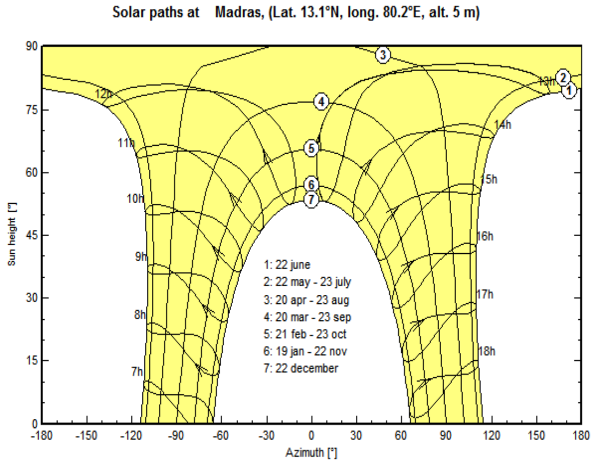

Fig. 3. Sketch of Solar Path

Solar panel selection - In photovoltaic systems, the preset is based on area or power. Depending on the size, the panel is selected based on the rated power and operating voltage. In the PV system, the choice of the frequency converter is based on the choice of the photovoltaic panel. The excess rating should match the panel's specifications.In the proposed work study, we simulated a 1 KW fixed tilt system that has been chosen to fix tilted linear shadows in Jaipur at (26.9124°N, 75.7873°E latitude and altitude.The time zone is selected according to the Indian Standard Time (IST).

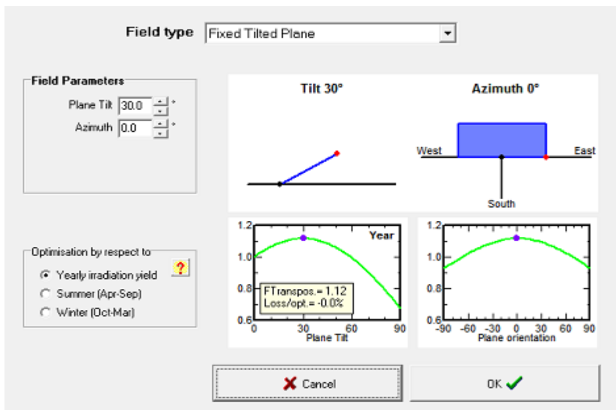

Fig. 4. Orientation and Horizon finalization of PV module

II. PARTIAL SHADING CONDITION

Partial shading is the case of poor performance of photovoltaic cells, panels or arrays, or provides low power,due to instability and complex IV and PV characteristics.In most cases, partial shadowing occurs when some of the PV cells on the panel or array block direct sunlight. Studies have shown that most shadows are caused by the angle of inclination of surrounding trees, clouds, buildings/houses, bird droppings, dust/leaves, water and solar panels/arrays. The full shadow will also bring similar problems to the PV system, but there is no discussion like partial shadows. Trees, nearby buildings and clouds are the main reasons for partial shadows.

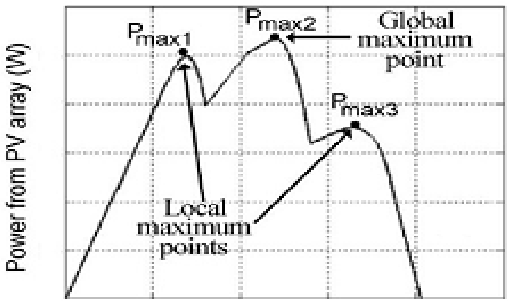

Fig. 5. P-V Characteristics of PV Array Under Partial Shading condition

Different causes (not appeared) can likewise influence the execution of the PV framework, including changing temperatures, climate conditions and daylight levels. The P-V attributes indicated can speak to a cluster under typical conditions. The P-V bend has various pinnacles, so it has different maximum power points (MPP), for this situation three. There is just a single worldwide MPP and two nearby MPPs, of which the worldwide MPP has the most elevated MPP esteem inside the P-V bend. MPP should be explored and broke down to check conceivable yields and to find MPPs all through the framework under explicit conditions. Hence, the MPPT algorithm must be connected to the framework to help track the MPP under all conditions, which will result in expanded yield and expanded effectiveness.

III. SIMULATION & RESULTS

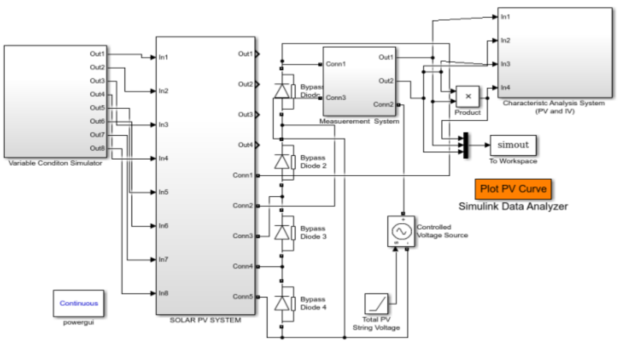

Fig. 6. Mathmeatical Model of Partial Shad ing Condition in Solar PV System

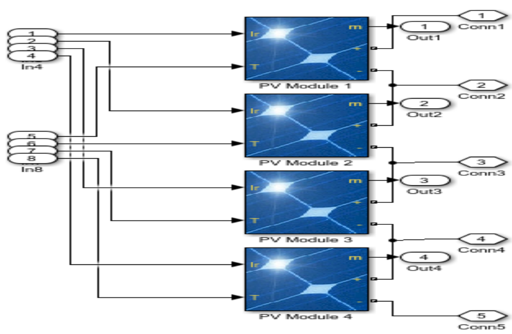

Fig. 7. Connection of PV String

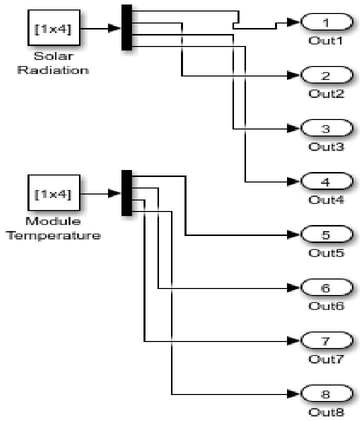

Fig. 8. Connection of Variable Irradiance and Variable temperature in PV String

A progressively broad incompletely shaded PV string with n distinctive irradi-ance estimations of G1,… ,Gn,, G1 > G2 > … > Gn, is partitioned into n sub-strings and their PV module quantities of their substrings are, separately, N1,… ,Nn. In light of the reenactment results introduced in this segment, a few valuable tenets for an incompletely shaded PV string when all is said in done case are divulged as the followings:

[1] The general V– I bend takes the state of a rising numerous progression patio field and V– P bend has mul-tiple tops, where the quantities of MPPs and step porch field are actually equivalent to the quantity of the irradiance levels in the PV string.

[2] The left-side MPP on the V– P bend is just reliant on sub-string1 comprised of full sun PV modules and autonomous on alternate substrings comprised of full shaded PV modules, which are skirted.

[3] At right-side MPP on the V– P bend, all modules in the string make commitment to power age, which is overwhelmed by the irradiance esteem Gn got in substring n, since the present estimation of Impn at the MPPn is relatively to Gn.

[4] The area of MPP is controlled by Gi and N1, … , Ni, since substring1 up to substring I make commitment to power age, the current Impi at the MPPi is relatively to Gi and alternate substrings from I +1 to n are avoided.

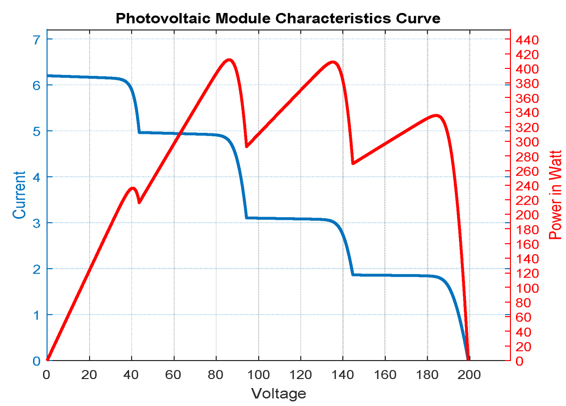

Fig. 9. P-V & I-V Characteristics of PV Array Under Partial Shading condition

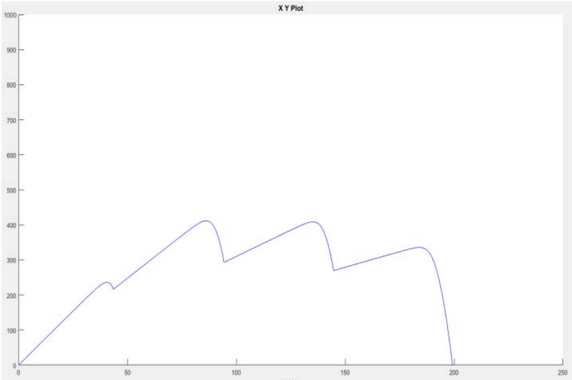

Fig. 10. P-V Characteristics of PV Array Under Partial Shading condition

Given a PV array consisting of N PV modules are arranged into Np PV module strings connected in parallel, each string with Ns PV modules in series, where N= Ns × Np. It is required to obtain the entire V–I and V–P characteristics curves for one to learn and under- stand the behaviour of a PV array in a complex scenario conclusion

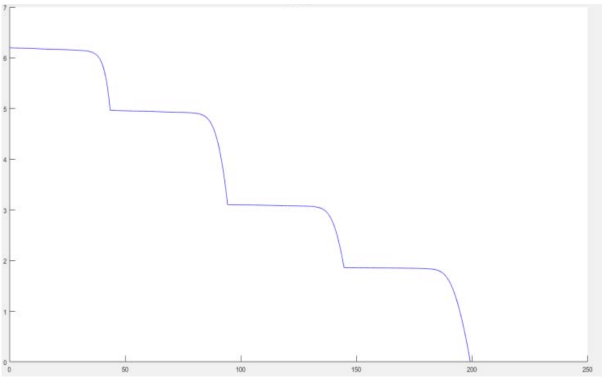

Fig. 11. I-V Characteristics of PV Array Under Partial Shading condition

IV. CONCLUSION

This paper initially presents the halfway shading condition investigation apparatuses like MATLAB and PV-Syst for analyzng incomplete shading condition. This paper center around the normal for photovoltaic framework. A mathematcal show has been inspected utilizing MATLAB to get to the impact of variable irradiance and variable temperature on PV and IV normal for solar photovoltiac system.This investigation is valuable in concentrate fractional shading condition impact on tracking maximum power point in such situation. This examination will help in use of MPPT algorithm in halfway shading scenatio for proficiency enhancement objective.

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